

# Command Line Interface Cheat Sheet

Windows CMD · PowerShell · Bash (Linux and macOS) · Git

**How to read this sheet.** Each row shows the same task in all three shells. Commands are in monospace; *grey italics* explain flags or name an alias for the same command.  needs administrator, root or sudo.  destructive; check before running. (Linux) or (macOS) marks a command that exists only on that platform. ★ marks the everyday essentials. Click any command to copy it.

## Everyday Essentials

The commands most people use every day, in short form. Click a task name for the full entry with more options and notes.

| Task                          | CMD                                                                                 | PowerShell                                                                                          | Bash                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| ★ <u>Change directory</u>     | cd C:\Users<br>cd ../Desktop                                                        | cd "C:\Program Files"<br>Set-Location ~\Desktop                                                     | cd /home/user/docs<br>cd ~/Desktop                                                 |
| ★ <u>List files</u>           | dir<br>dir /A <i>(include hidden)</i>                                               | Get-ChildItem <i>(alias: ls, dir, gci)</i><br>ls -Force <i>(include hidden)</i>                     | ls<br>ls -la <i>(long, incl. hidden)</i>                                           |
| ★ <u>Copy files</u>           | copy file.txt C:\Backup<br>copy *.txt D:\Backup                                     | Copy-Item file.txt ~\Backup <i>(alias: cp)</i><br>Copy-Item *.txt -Destination C:\Backup            | cp file.txt ~/Backup<br>cp -i file.txt ~/Backup <i>(prompt before overwrite)</i>   |
| ★ <u>Move or rename</u>       | move file.txt C:\New<br>ren old.txt new.txt                                         | Move-Item file.txt C:\New <i>(alias: mv)</i><br>Rename-Item old.txt new.txt                         | mv file.txt ~/new/<br>mv old.txt new.txt                                           |
| ★ <u>Delete files</u>         | del oldfile.txt<br>del *.tmp                                                        | Remove-Item oldfile.txt <i>(alias: rm, del)</i><br>Remove-Item *.tmp -WhatIf <i>(dry run)</i>       | rm oldfile.txt<br>rm -i file.txt <i>(confirm)</i>                                  |
| ★ <u>View file contents</u>   | type readme.txt                                                                     | Get-Content readme.txt <i>(alias: cat, type, gc)</i>                                                | cat readme.txt<br>less readme.txt <i>(paginated; q to quit)</i>                    |
| ★ <u>Find files</u>           | dir /S /B *.txt<br>where /R C:\Projects *.log                                       | Get-ChildItem -Recurse -Filter *.txt<br>gci -r -Include *.log,*.txt                                 | find . -name "*.txt"<br>find /home -type f -name "*.log"                           |
| ★ <u>Search text in files</u> | findstr "error" log.txt<br>findstr /S /I "text" *.* <i>(recursive, ignore case)</i> | Select-String -Path log.txt -Pattern error<br><i>(alias: sls)</i><br>sls error *.log -CaseSensitive | grep "error" log.txt<br>grep -ri "error" /var/log/ <i>(recursive, ignore case)</i> |
| ★ <u>List processes</u>       | tasklist<br>tasklist /V <i>(verbose)</i>                                            | Get-Process <i>(alias: ps, gps)</i><br>Get-Process -Name chrome                                     | ps aux<br>ps aux   grep nginx                                                      |
| ★ <u>Kill process</u>         | taskkill /IM notepad.exe<br>taskkill /PID 1234 /F <i>(force)</i>                    | Stop-Process -Name notepad <i>(alias: kill)</i><br>Stop-Process -Id 1234 -Force                     | kill 1234 <i>(SIGTERM, polite)</i><br>kill -9 1234 <i>(SIGKILL, force)</i>         |

| Task                                | CMD                                                          | PowerShell                                                                                                         | Bash                                                                                     |
|-------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| ★ <u>Ping host</u>                  | ping google.com <i>(4 packets)</i><br>ping -n 10 192.168.1.1 | Test-Connection google.com<br>Test-Connection 8.8.8.8 -Count 5                                                     | ping google.com <i>(until Ctrl+C)</i><br>ping -c 5 google.com                            |
| ★ <u>IP configuration</u>           | ipconfig<br>ipconfig /all <i>(MAC, DNS, DHCP)</i>            | Get-NetIPConfiguration<br>Get-NetIPAddress -AddressFamily IPv4                                                     | ip addr <i>(Linux; short form: ip a)</i><br>ifconfig <i>(macOS; deprecated on Linux)</i> |
| ★ <u>Show environment variables</u> | set<br>set PATH <i>(all starting with PATH)</i>              | Get-ChildItem Env: <i>(or: dir env:)</i><br>\$env:PATH                                                             | printenv<br>env                                                                          |
| ★ <u>Command history</u>            | doskey /history<br>F7 <i>(popup list)</i>                    | Get-History <i>(alias: h, history)</i><br>Get-Content (Get-PSReadLineOption).HistorySavePath <i>(all sessions)</i> | history<br>history 20 <i>(last 20)</i>                                                   |
| ★ <u>Get help</u>                   | dir /?<br>help copy                                          | Get-Help Get-Process -Examples<br>Get-Help *-Service <i>(search by name)</i>                                       | man ls<br>ls --help                                                                      |

## File and Directory Management

| Task                          | CMD                                                                                                                                | PowerShell                                                                                                            | Bash                                                                                                                                   | Notes                                                                                                                         |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| ★ <u>Change directory</u>     | cd C:\Users<br>cd ..\Desktop<br>cd /d D:\Data <i>(switch drive)</i><br>D: <i>(switch drive only)</i>                               | cd "C:\Program Files"<br>Set-Location ~\Desktop<br>cd - <i>(previous dir, PS 6+)</i>                                  | cd /home/user/docs<br>cd ~/Desktop<br>cd - <i>(previous dir)</i><br>cd <i>(home)</i>                                                   | cd .. goes to the parent. ~ is the home directory in PowerShell and Bash. CMD needs /d to change drive and directory at once. |
| ★ <u>List files</u>           | dir<br>dir /A <i>(include hidden)</i><br>dir /S <i>(recursive)</i><br>dir /B <i>(bare names)</i><br>dir /O-D <i>(newest first)</i> | Get-ChildItem <i>(alias: ls, dir, gci)</i><br>ls -Force <i>(include hidden)</i><br>Get-ChildItem -Recurse<br>ls *.txt | ls<br>ls -la <i>(long, incl. hidden)</i><br>ls -lh <i>(readable sizes)</i><br>ls -lt <i>(newest first)</i><br>ls -R <i>(recursive)</i> |                                                                                                                               |
| <u>Show current directory</u> | cd <i>(no arguments)</i><br>echo %CD%                                                                                              | Get-Location <i>(alias: pwd)</i><br>\$PWD                                                                             | pwd                                                                                                                                    |                                                                                                                               |
| <u>Create directory</u>       | mkdir MyReports<br>md Folder1\Folder2 <i>(nested is fine)</i>                                                                      | mkdir ~/Projects/New<br>New-Item -ItemType Directory MyReports                                                        | mkdir MyReports<br>mkdir -p a/b/c <i>(nested)</i>                                                                                      |                                                                                                                               |
| <u>Remove directory</u>       | rmdir OldFolder <i>(empty only)</i><br>rmdir /S /Q OldFolder <i>(with contents, no prompt)</i>                                     | Remove-Item OldFolder -Recurse -Force<br>Remove-Item OldFolder -Recurse -WhatIf <i>(preview)</i>                      | rmdir OldFolder <i>(empty only)</i><br>rm -r OldFolder<br>rm -rf OldFolder <i>(no prompts)</i>                                         | ⚠ Recursive deletes bypass the Recycle Bin / Trash. There is no undo.                                                         |
| <u>Directory tree</u>         | tree C:\Projects<br>tree /F <i>(include files)</i>                                                                                 | tree /F <i>(same tree.com as CMD)</i><br>Get-ChildItem -Recurse -Name                                                 | tree ~/Projects<br>tree -L 2 <i>(limit depth)</i>                                                                                      | tree is not preinstalled on most Linux distros or macOS: apt install tree, brew install tree.                                 |

| Task                   | CMD                                                                                                                                     | PowerShell                                                                                                                     | Bash                                                                                                                                | Notes                                                                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ Copy files           | copy file.txt C:\Backup<br>copy *.txt D:\Backup                                                                                         | Copy-Item file.txt ~\Backup <i>(alias: cp)</i><br>Copy-Item *.txt -Destination C:\Backup                                       | cp file.txt ~/Backup<br>cp -i file.txt ~/Backup <i>(prompt before overwrite)</i>                                                    |                                                                                                                                                    |
| Copy directories       | xcopy C:\Src D:\Dst /E /I<br>robocopy C:\Src D:\Dst /E<br>robocopy C:\Src D:\Dst /MIR <i>(mirror; deletes extras)</i>                   | Copy-Item folder -Recurse -Destination dest                                                                                    | cp -r folder/ ~/dest<br>cp -a folder/ ~/dest <i>(preserve attributes)</i><br>rsync -av src/ dest/                                   | robocopy and rsync are the robust choices for large copies: both can resume and mirror. ⚠ /MIR and rsync --delete remove files at the destination. |
| ★ Move or rename       | move file.txt C:\New<br>ren old.txt new.txt                                                                                             | Move-Item file.txt C:\New <i>(alias: mv)</i><br>Rename-Item old.txt new.txt                                                    | mv file.txt ~/new/<br>mv old.txt new.txt                                                                                            |                                                                                                                                                    |
| ★ Delete files         | del oldfile.txt<br>del *.tmp<br>del /P file.txt <i>(prompt)</i>                                                                         | Remove-Item oldfile.txt <i>(alias: rm, del)</i><br>Remove-Item *.tmp -WhatIf <i>(dry run)</i>                                  | rm oldfile.txt<br>rm -i file.txt <i>(confirm)</i><br>rm *.tmp                                                                       | ⚠ No Recycle Bin. In PowerShell, -WhatIf previews any destructive cmdlet.                                                                          |
| ★ View file contents   | type readme.txt                                                                                                                         | Get-Content readme.txt <i>(alias: cat, type, gc)</i>                                                                           | cat readme.txt<br>less readme.txt <i>(paginated; q to quit)</i>                                                                     |                                                                                                                                                    |
| Create empty file      | type nul > notes.txt<br>copy con notes.txt <i>(type text, then Ctrl+Z, Enter)</i>                                                       | New-Item -ItemType File notes.txt <i>(alias: ni)</i>                                                                           | touch notes.txt<br>> notes.txt                                                                                                      | touch also updates the timestamp of an existing file. New-Item errors if the file exists unless you add -Force.                                    |
| ★ Find files           | dir /S /B *.txt<br>where /R C:\Projects *.log                                                                                           | Get-ChildItem -Recurse -Filter *.txt<br>gci -r -Include *.log,*.txt<br>gci -r   Where LastWriteTime -gt (Get-Date).AddDays(-1) | find . -name "*.txt"<br>find /home -type f -name "*.log"<br>find . -mtime -1 <i>(modified in last day)</i><br>find . -size +100M    |                                                                                                                                                    |
| Which command runs     | where python                                                                                                                            | Get-Command python<br>Get-Command ls <i>(shows what an alias points to)</i>                                                    | which python<br>type ls <i>(shows aliases and builtins too)</i><br>command -v python                                                |                                                                                                                                                    |
| Compare files          | fc file1.txt file2.txt<br>fc /B a.bin b.bin <i>(binary)</i>                                                                             | Compare-Object (Get-Content a.txt) (Get-Content b.txt) <i>(alias: diff)</i>                                                    | diff file1.txt file2.txt<br>diff -u a.txt b.txt <i>(unified format)</i>                                                             | PowerShell's diff alias is not Unix diff: it compares sets of objects and ignores line order unless you add -SyncWindow 0.                         |
| ★ Search text in files | findstr "error" log.txt<br>findstr /S /I "text" *.* <i>(recursive, ignore case)</i><br>findstr /N "text" file.txt <i>(line numbers)</i> | Select-String -Path log.txt -Pattern error <i>(alias: sls)</i><br>sls error *.log -CaseSensitive<br>gci -r *.log   sls error   | grep "error" log.txt<br>grep -ri "error" /var/log/ <i>(recursive, ignore case)</i><br>grep -n "text" file.txt <i>(line numbers)</i> |                                                                                                                                                    |

| Task                  | CMD                                                                                                                      | PowerShell                                                                                       | Bash                                                                                                                    | Notes                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| File permissions      | icacls file.txt<br>icacls file.txt /grant User:F                                                                         | Get-Acl file.txt<br>Set-Acl file.txt -AclObject \$acl<br>icacls file.txt <i>(works here too)</i> | chmod 755 script.sh<br>chmod +x script.sh<br>chown user:group file.txt                                                  | 🔒 Changing permissions on files you do not own needs admin or sudo. Windows uses ACLs; Unix uses rwx bits.            |
| File attributes       | attrib +r file.txt <i>(read-only)</i><br>attrib +h file.txt <i>(hidden)</i><br>attrib -r -h file.txt                     | Set-ItemProperty file.txt IsReadOnly \$true<br>(Get-Item file.txt).Attributes                    | chattr +i file.txt <i>(immutable, Linux)</i><br>chattr -i file.txt<br>chflags hidden file.txt <i>(macOS)</i>            | Unix has no hidden attribute: files whose names start with a dot are hidden by convention.                            |
| Symbolic links        | mklink link.txt C:\target.txt<br>mklink /D linkdir C:\target                                                             | New-Item -ItemType SymbolicLink -Path link -Target target                                        | ln -s target link<br>ln -s /path/to/dir linkdir                                                                         | 🔒 Windows needs an admin prompt or Developer Mode to create symlinks. Bash argument order is target first, then link. |
| Compress / archive    | tar -a -cf archive.zip folder <i>(Windows 10 1803+)</i><br>compact /C file.txt <i>(NTFS compression, not an archive)</i> | Compress-Archive -Path folder -DestinationPath archive.zip                                       | tar -czvf archive.tar.gz folder/<br>zip -r archive.zip folder/<br>gzip file.txt <i>(replaces file with file.txt.gz)</i> | bsdtar ships with Windows 10 and later and works in both CMD and PowerShell. It reads .zip, .tar.gz, .7z and more.    |
| Extract               | tar -xf archive.zip<br>tar -xf archive.tar.gz<br>expand archive.cab C:\dest <i>(CAB only)</i>                            | Expand-Archive archive.zip -DestinationPath dest\                                                | unzip archive.zip<br>tar -xzf archive.tar.gz<br>gunzip file.txt.gz                                                      |                                                                                                                       |
| File hash / checksum  | certutil -hashfile file.iso SHA256                                                                                       | Get-FileHash file.iso <i>(SHA256 by default)</i><br>Get-FileHash file.iso -Algorithm MD5         | sha256sum file.iso<br>md5sum file.iso<br>shasum -a 256 file.iso <i>(macOS)</i>                                          |                                                                                                                       |
| Open with default app | start report.pdf<br>start . <i>(Explorer here)</i><br>start https://example.com                                          | Invoke-Item report.pdf <i>(alias: ii)</i><br>start . <i>(alias of Start-Process)</i>             | xdg-open report.pdf <i>(Linux)</i><br>open report.pdf <i>(macOS)</i><br>open . <i>(Finder here, macOS)</i>              |                                                                                                                       |
| Clear screen          | cls                                                                                                                      | Clear-Host <i>(alias: cls, clear)</i><br>Ctrl+L                                                  | clear<br>Ctrl+L                                                                                                         |                                                                                                                       |

## Disk and Storage

| Task            | CMD                                                                                               | PowerShell                                                                                                                | Bash                                         | Notes                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------|
| Free disk space | fsutil volume diskfree C:<br>wmic logicaldisk get name,size,freespace <i>(wmic is deprecated)</i> | Get-PSDrive -PSProvider FileSystem<br>Get-Volume<br>Get-CimInstance Win32_LogicalDisk  <br>Select DeviceID,Size,FreeSpace | df -h<br>df -h /home <i>(one filesystem)</i> | wmic is deprecated and removed from recent Windows 11 builds. Prefer Get-CimInstance. |

| Task                    | CMD                                                                                                                      | PowerShell                                                                                                                          | Bash                                                                                      | Notes                                                                                       |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Disk and partition info | diskpart <i>(interactive: list disk, list volume)</i><br>wmic diskdrive get model,size<br><i>(deprecated)</i>            | Get-Disk<br>Get-Partition<br>Get-Volume                                                                                             | lsblk <i>(Linux)</i><br>sudo fdisk -l <i>(Linux)</i><br>diskutil list <i>(macOS)</i>      |                                                                                             |
| Folder size             | dir /S C:\folder <i>(total at the end)</i>                                                                               | (Get-ChildItem -Recurse folder   Measure-Object Length -Sum).Sum / 1MB                                                              | du -sh folder/<br>du -h --max-depth=1 .   sort -h<br><i>(Linux)</i><br>du -sh *   sort -h |                                                                                             |
| Check disk              | chkdsk C:<br>chkdsk C: /F <i>(fix errors; system drive needs a reboot)</i>                                               | Repair-Volume -DriveLetter C -Scan<br>Repair-Volume -DriveLetter C -OfflineScanAndFix                                               | sudo fsck /dev/sda1 <i>(unmount first)</i><br>diskutil verifyVolume / <i>(macOS)</i>      | 🔒 Requires admin or sudo.<br>Never run fsck on a mounted filesystem.                        |
| Format drive            | format D: /FS:NTFS /Q                                                                                                    | Format-Volume -DriveLetter D -FileSystem NTFS                                                                                       | sudo mkfs.ext4 /dev/sdb1<br>diskutil eraseDisk APFS Name disk2<br><i>(macOS)</i>          | ⚠️⚠️ Destroys everything on the volume. Double-check the drive letter or device name first. |
| Mount / map drives      | net use Z: \\server\share <i>(map network drive)</i><br>net use Z: /delete<br>mountvol <i>(list volume mount points)</i> | New-PSDrive -Name Z -PSProvider FileSystem -Root \\server\share -Persist<br>Mount-DiskImage file.iso<br>Dismount-DiskImage file.iso | sudo mount /dev/sdb1 /mnt/usb<br>sudo umount /mnt/usb<br>mount <i>(list mounts)</i>       |                                                                                             |

## Windows Repair

| Task                              | Command                                                                                                                                                                                                                | Notes                                                                                                                                                                                                                     |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repair system files (SFC)         | sfc /scannow <i>(scan and repair protected system files)</i><br>sfc /verifyonly <i>(scan only, no repair)</i><br>sfc /scannow /offbootdir=C:\ /offwindir=C:\Windows <i>(offline, from recovery)</i>                    | 🔒 Run from an elevated prompt. If SFC reports files it could not fix, run the DISM RestoreHealth step below, then run sfc /scannow again.                                                                                 |
| Repair the component store (DISM) | DISM /Online /Cleanup-Image /CheckHealth <i>(quick flag check)</i><br>DISM /Online /Cleanup-Image /ScanHealth <i>(full scan, no repair)</i><br>DISM /Online /Cleanup-Image /RestoreHealth <i>(download and repair)</i> | 🔒 Elevated. RestoreHealth fixes the store SFC copies from, so run it first when files are corrupt. It uses Windows Update; add /Source:wim:X:\sources\install.wim:1 /LimitAccess to repair from an install image offline. |
| Check disk (CHKDSK)               | chkdsk C: <i>(read-only report)</i><br>chkdsk C: /F <i>(fix file-system errors)</i><br>chkdsk C: /R <i>(find bad sectors, implies /F)</i>                                                                              | 🔒 Elevated. The system drive cannot be locked while Windows is running, so the check is scheduled for the next reboot. Also listed under Disk and Storage.                                                                |
| System Restore                    | rstrui <i>(open the System Restore wizard)</i><br>Get-ComputerRestorePoint <i>(PowerShell: list restore points)</i><br>Checkpoint-Computer -Description "Before change" <i>(PowerShell: create one)</i>                | 🔒 Creating or restoring needs admin. System Protection must be turned on for the drive first (System Properties > System Protection).                                                                                     |

| Task                     | Command                                                                                                               | Notes                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Boot to recovery options | shutdown /r /o /t 0 <i>(reboot into Advanced Startup)</i><br>reagentc /info <i>(show recovery environment status)</i> | Advanced Startup leads to Startup Repair, Safe Mode, System Restore and a command prompt. Or hold Shift while clicking Restart. |

## System and User Information

| Task               | CMD                                                                                           | PowerShell                                                                                                                                                         | Bash                                                                                                        | Notes                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Current user       | whoami<br>echo %USERNAME%<br>whoami /groups                                                   | whoami<br>\$env:USERNAME<br>[Environment]::UserName                                                                                                                | whoami<br>echo \$USER<br>id <i>(uid, gid and groups)</i>                                                    |                                                                                 |
| Hostname           | hostname<br>echo %COMPUTERNAME%                                                               | hostname<br>\$env:COMPUTERNAME                                                                                                                                     | hostname<br>hostnamectl <i>(Linux with systemd)</i><br>scutil --get ComputerName <i>(macOS)</i>             |                                                                                 |
| System information | systeminfo<br>systeminfo   findstr /B /C:"OS Name" /C:"OS Version"                            | Get-ComputerInfo<br>Get-ComputerInfo   Select<br>OsName,OsVersion,CsTotalPhysicalMemory<br>Get-CimInstance Win32_OperatingSystem                                   | uname -a<br>cat /etc/os-release <i>(Linux)</i><br>hostnamectl <i>(Linux)</i><br>sw_vers <i>(macOS)</i>      | Get-ComputerInfo is slow (several seconds). Pick properties with Select-Object. |
| OS version         | ver                                                                                           | [Environment]::OSVersion<br>\$PSVersionTable.PSVersion <i>(PowerShell version)</i>                                                                                 | uname -r <i>(kernel)</i><br>lsb_release -a <i>(Debian/Ubuntu)</i><br>sw_vers -productVersion <i>(macOS)</i> |                                                                                 |
| CPU                | echo %NUMBER_OF_PROCESSORS%<br>wmic cpu get name,numberofcores<br><i>(deprecated)</i>         | Get-CimInstance Win32_Processor  <br>Select<br>Name,NumberOfCores,NumberOfLogicalProc<br>essors                                                                    | lscpu <i>(Linux)</i><br>nproc <i>(Linux)</i><br>sysctl -n machdep.cpu.brand_string<br><i>(macOS)</i>        |                                                                                 |
| Memory             | systeminfo   findstr /C:"Total Physical Memory" /C:"Available Physical Memory"                | Get-CimInstance Win32_OperatingSystem   Select<br>TotalVisibleMemorySize,FreePhysicalMem<br>ory<br>Get-CimInstance Win32_PhysicalMemory  <br>Select Capacity,Speed | free -h <i>(Linux)</i><br>cat /proc/meminfo <i>(Linux)</i><br>vm_stat <i>(macOS)</i><br>top <i>(any)</i>    |                                                                                 |
| Uptime             | systeminfo   findstr /C:"System Boot Time"<br>net statistics workstation  <br>findstr "since" | (Get-Date) - (Get-CimInstance<br>Win32_OperatingSystem).LastBootUpTime<br>Get-Uptime <i>(PowerShell 6+ only)</i>                                                   | uptime<br>uptime -p <i>(pretty, Linux)</i><br>uptime -s <i>(boot time, Linux)</i>                           |                                                                                 |

| Task            | CMD                                      | PowerShell                                                                         | Bash                                                               | Notes                                                                              |
|-----------------|------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Date and time   | date /T<br>time /T<br>echo %DATE% %TIME% | Get-Date<br>Get-Date -Format "yyyy-MM-dd HH:mm:ss"<br>Get-Date -UFormat "%Y-%m-%d" | date<br>date +"%Y-%m-%d %H:%M:%S"<br>date -u (UTC)                 | Without /T, CMD's date and time commands prompt you to change the clock (🔒 admin). |
| Logged-in users | query user (alias: quser)                | query user<br>Get-CimInstance Win32_ComputerSystem  <br>Select UserName            | who<br>w (who, plus what they are running)<br>last (login history) |                                                                                    |

## Processes, Services and Power

| Task                      | CMD                                                                                                               | PowerShell                                                                                                                             | Bash                                                                                                                         | Notes                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ★ List processes          | tasklist<br>tasklist /V (verbose)<br>tasklist /FI "IMAGENAME eq chrome.exe"                                       | Get-Process (alias: ps, gps)<br>Get-Process -Name chrome<br>Get-Process   Sort-Object CPU -Descending   Select -First 10               | ps aux<br>ps aux   grep nginx<br>top (live)<br>htop (live, nicer; needs install)                                             |                                                                                               |
| ★ Kill process            | taskkill /IM notepad.exe<br>taskkill /PID 1234 /F (force)<br>taskkill /F /IM chrome.exe /T (with child processes) | Stop-Process -Name notepad (alias: kill)<br>Stop-Process -Id 1234 -Force                                                               | kill 1234 (SIGTERM, polite)<br>kill -9 1234 (SIGKILL, force)<br>pkill -f pattern<br>killall firefox                          | ⚠ Force-killing skips cleanup and can corrupt files being written. Try the polite form first. |
| Find process using a port | netstat -ano   findstr :8080 (PID in last column)                                                                 | Get-NetTCPConnection -LocalPort 8080  <br>Select OwningProcess<br>Get-Process -Id (Get-NetTCPConnection -LocalPort 8080).OwningProcess | sudo lsof -i :8080<br>sudo ss -ltnp   grep 8080 (Linux)                                                                      |                                                                                               |
| Run in background         | start /B program.exe (same window)<br>start "" notepad.exe (new window)                                           | Start-Process notepad<br>Start-Job { long-task }; Get-Job;<br>Receive-Job 1<br>command & (PowerShell 6+)                               | command &<br>nohup command & (survives logout)<br>jobs / fg / bg<br>Ctrl+Z (suspend the foreground job)                      |                                                                                               |
| List services             | sc query (running)<br>sc query state= all<br>net start (running only)                                             | Get-Service<br>Get-Service   Where Status -eq Running<br>Get-Service -Name Spooler                                                     | systemctl list-units --type=service (Linux)<br>systemctl list-units --type=service --state=running<br>launchctl list (macOS) |                                                                                               |

| Task                                  | CMD                                                                                                                 | PowerShell                                                                                                                       | Bash                                                                                                                                          | Notes                                                                                                                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Start / stop / restart service</b> | sc start Spooler<br>sc stop Spooler<br>net stop Spooler && net start Spooler<br><i>(net start/stop also work)</i>   | Start-Service Spooler<br>Stop-Service Spooler<br>Restart-Service Spooler                                                         | sudo systemctl start nginx<br>sudo systemctl stop nginx<br>sudo systemctl restart nginx<br>sudo systemctl enable nginx <i>(start at boot)</i> |  Requires admin or sudo. sc and Get-Service use the service name (Spooler), not the display name (Print Spooler). |
| <b>Service status</b>                 | sc query Spooler<br>sc qc Spooler <i>(configuration)</i>                                                            | Get-Service Spooler<br>(Get-Service Spooler).Status                                                                              | systemctl status nginx<br>systemctl is-active nginx<br>journalctl -u nginx -f <i>(follow logs, Linux)</i>                                     |                                                                                                                                                                                                      |
| <b>Scheduled tasks</b>                | schtasks /query<br>schtasks /create /tn Backup /tr C:\backup.bat /sc daily /st 02:00<br>schtasks /delete /tn Backup | Get-ScheduledTask<br>Get-ScheduledTask Backup   Start-ScheduledTask<br>Register-ScheduledTask <i>(see the Get-Help examples)</i> | crontab -l <i>(list)</i><br>crontab -e <i>(edit)</i><br>0 2 * * * /path/backup.sh <i>(daily at 02:00)</i><br>launchctl <i>(macOS launchd)</i> |                                                                                                                                                                                                      |
| <b>Restart system</b>                 | shutdown /r /t 0<br>shutdown /a <i>(abort a pending shutdown)</i>                                                   | Restart-Computer<br>Restart-Computer -Force                                                                                      | sudo reboot<br>sudo shutdown -r now<br>sudo shutdown -r +5 <i>(in 5 minutes)</i>                                                              |  Requires admin or sudo.                                                                                          |
| <b>Shut down system</b>               | shutdown /s /t 0<br>shutdown /s /t 60 <i>(in 60 seconds)</i><br>shutdown /h <i>(hibernate)</i>                      | Stop-Computer<br>Stop-Computer -Force                                                                                            | sudo shutdown -h now<br>sudo poweroff<br>sudo shutdown -c <i>(cancel, Linux)</i>                                                              |  Requires admin or sudo.                                                                                          |

## Networking and Diagnostics

| Task                              | CMD                                                                                                                                                                                              | PowerShell                                                                                                        | Bash                                                                                                                                   | Notes                                                                                         |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ★ <b>Ping host</b>                | ping google.com <i>(4 packets)</i><br>ping -n 10 192.168.1.1<br>ping -t host <i>(until Ctrl+C)</i>                                                                                               | Test-Connection google.com<br>Test-Connection 8.8.8.8 -Count 5<br>Test-Connection host -Quiet <i>(True/False)</i> | ping google.com <i>(until Ctrl+C)</i><br>ping -c 5 google.com                                                                          | Windows ping stops after 4 packets by default. Unix ping runs until you stop it.              |
| ★ <b>IP configuration</b>         | ipconfig<br>ipconfig /all <i>(MAC, DNS, DHCP)</i><br>ipconfig /release && ipconfig /renew                                                                                                        | Get-NetIPConfiguration<br>Get-NetIPAddress -AddressFamily IPv4                                                    | ip addr <i>(Linux; short form: ip a)</i><br>ifconfig <i>(macOS; deprecated on Linux)</i><br>hostname -I <i>(Linux, addresses only)</i> |                                                                                               |
| <b>Public IP</b>                  | curl ifconfig.me                                                                                                                                                                                 | Invoke-RestMethod ifconfig.me                                                                                     | curl ifconfig.me                                                                                                                       |                                                                                               |
| <b>Connections and open ports</b> | netstat -an <i>(all)</i><br>netstat -ano <i>(with PIDs)</i><br>netstat -b <i>(with program names,  admin)</i> | Get-NetTCPConnection<br>Get-NetTCPConnection -State Listen                                                        | ss -tulpn <i>(Linux, preferred)</i><br>sudo netstat -tulpn <i>(older Linux)</i><br>sudo lsof -i -P <i>(macOS and Linux)</i>            | netstat and ifconfig are deprecated on Linux in favour of ss and ip, but still work on macOS. |

| Task                | CMD                                                                                                                                                                | PowerShell                                                                                                                                       | Bash                                                                                                                                          | Notes                                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| DNS lookup          | nslookup google.com<br>nslookup google.com 8.8.8.8 <i>(ask a specific server)</i><br>nslookup -type=MX google.com                                                  | Resolve-DnsName google.com<br>Resolve-DnsName google.com -Type MX<br>Resolve-DnsName google.com -Server 8.8.8.8                                  | dig google.com<br>dig +short google.com<br>dig MX google.com<br>host google.com                                                               |                                                                                                       |
| Trace route         | tracert google.com<br>tracert -d google.com <i>(skip DNS, faster)</i><br>pathping google.com                                                                       | Test-NetConnection google.com -TraceRoute                                                                                                        | tracert google.com<br>tracert google.com <i>(Linux, no root needed)</i><br>mtr google.com <i>(live; needs install)</i>                        |                                                                                                       |
| Routing table       | route print<br>route add 192.168.2.0 mask 255.255.255.0 192.168.1.1                                                                                                | Get-NetRoute<br>New-NetRoute -DestinationPrefix 192.168.2.0/24 -NextHop 192.168.1.1 -InterfaceIndex 12                                           | ip route <i>(Linux)</i><br>sudo ip route add 192.168.2.0/24 via 192.168.1.1<br>netstat -rn <i>(macOS)</i>                                     | 🔒 Adding routes needs admin or sudo.                                                                  |
| Flush DNS cache     | ipconfig /flushdns<br>ipconfig /displaydns                                                                                                                         | Clear-DnsClientCache<br>Get-DnsClientCache                                                                                                       | sudo resolvectl flush-caches <i>(Linux, systemd-resolved)</i><br>sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder <i>(macOS)</i> |                                                                                                       |
| ARP table           | arp -a                                                                                                                                                             | Get-NetNeighbor<br>Get-NetNeighbor -State Reachable                                                                                              | ip neigh <i>(Linux)</i><br>arp -a <i>(macOS and Linux)</i>                                                                                    |                                                                                                       |
| Download file       | curl -O https://example.com/file.zip<br>curl -L -o out.zip URL <i>(follow redirects, custom name)</i><br>certutil -urlcache -split -f URL file.zip <i>(legacy)</i> | Invoke-WebRequest URL -OutFile file.zip <i>(alias: iwr)</i><br>Invoke-RestMethod URL <i>(parses JSON)</i> <i>(alias: irm)</i><br>curl.exe -O URL | wget URL<br>curl -O URL<br>curl -L -o out.zip URL                                                                                             | In Windows PowerShell 5.1, curl is an alias of Invoke-WebRequest. Type curl.exe to get the real tool. |
| Test a port         | curl -v telnet://host:443<br>telnet host 80 <i>(optional feature, off by default)</i>                                                                              | Test-NetConnection host -Port 443 <i>(alias: tnc)</i><br>tnc host -Port 443 -InformationLevel Quiet                                              | nc -zv host 443<br>curl -v telnet://host:443 <i>(echo &gt; /dev/tcp/host/443) &amp;&amp; echo open</i> <i>(Bash only)</i>                     |                                                                                                       |
| SSH and remote copy | ssh user@host<br>scp file.txt user@host:/path/<br>ssh -i key.pem user@host                                                                                         | ssh user@host<br>scp file.txt user@host:/path/<br>Enter-PSSession -ComputerName host <i>(WinRM, Windows to Windows)</i>                          | ssh user@host<br>scp file.txt user@host:/path/<br>rsync -avz src/ user@host:/dest/                                                            | The OpenSSH client is built into Windows 10 1809 and later.                                           |

| Task                                | CMD                                                                                                                                                                                    | PowerShell                                                                                                                                                           | Bash                                                                                                                                                                                                                        | Notes                                                                                                                                                                       |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wi-Fi profiles and passwords</b> | netsh wlan show profiles <i>(saved networks)</i><br>netsh wlan show profile name="MyWiFi" key=clear <i>(reveals the password)</i><br>netsh wlan connect name="MyWiFi"                  | netsh wlan show profiles<br>netsh wlan show profile name="MyWiFi" key=clear   sls Key <i>(just the password line)</i>                                                | nmcli device wifi list <i>(Linux)</i><br>sudo nmcli -s -g 802-11-wireless-security.psk connection show "MyWiFi" <i>(Linux password)</i><br>security find-generic-password -wa "MyWiFi" <i>(macOS; prompts for keychain)</i> | The Windows password shows as "Key Content" in the output. On Linux the psk read needs sudo; macOS prompts for your login keychain.                                         |
| <b>Firewall</b>                     | netsh advfirewall show allprofiles<br>netsh advfirewall firewall add rule name="App" dir=in action=allow program="C:\app.exe" enable=yes<br>netsh advfirewall set allprofiles state on | Get-NetFirewallProfile<br>New-NetFirewallRule -DisplayName "App" -Direction Inbound -Action Allow -Program "C:\app.exe"<br>Set-NetFirewallProfile -All -Enabled True | sudo ufw status<br>sudo ufw allow 22/tcp <i>(Debian/Ubuntu)</i><br>sudo firewall-cmd --add-port=8080/tcp --permanent <i>(Fedora/RHEL)</i><br>sudo pfctl -sr <i>(macOS rules)</i>                                            | 🔒 Requires admin or sudo. ufw is the friendly front end on Ubuntu; firewalld (firewall-cmd) on Fedora/RHEL; macOS uses pf plus the Application Firewall in System Settings. |

## Environment, History and Shell Basics

| Task                                | CMD                                                                                                                                                               | PowerShell                                                                                                                      | Bash                                                                                                                                                         | Notes                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| ★ <b>Show environment variables</b> | set<br>set PATH <i>(all starting with PATH)</i><br>echo %PATH%                                                                                                    | Get-ChildItem Env: <i>(or: dir env:)</i><br>\$env:PATH<br>\$env:PATH -split ';' <i>(one per line)</i>                           | printenv<br>env<br>echo \$PATH<br>echo \$PATH   tr ':' '\n' <i>(one per line)</i>                                                                            |                                                         |
| <b>Set environment variable</b>     | set MYVAR=value <i>(this session)</i><br>setx MYVAR value <i>(permanent, new windows only)</i><br>setx PATH "%PATH%;C:\tools" ⚠️ <i>(truncates at 1024 chars)</i> | \$env:MYVAR = 'value' <i>(this session)</i><br>[Environment]::SetEnvironmentVariable('MYVAR','value','User') <i>(permanent)</i> | export MYVAR=value <i>(this session)</i><br>MYVAR=value command <i>(one command only)</i><br>add the export line to ~/.bashrc or ~/.zshrc <i>(permanent)</i> | Permanent changes only affect shells opened afterwards. |
| ★ <b>Command history</b>            | doskey /history<br>F7 <i>(popup list)</i>                                                                                                                         | Get-History <i>(alias: h, history)</i><br>Get-Content (Get-PSReadLineOption).HistorySavePath <i>(all sessions)</i>              | history<br>history 20 <i>(last 20)</i><br>history   grep ssh                                                                                                 |                                                         |
| <b>Re-run a command</b>             | F3 or ↑ <i>(previous)</i>                                                                                                                                         | ↑ <i>(previous)</i><br>Invoke-History 5 <i>(alias: r 5)</i>                                                                     | !! <i>(previous)</i><br>!5 <i>(history entry 5)</i><br>!ssh <i>(last command starting with ssh)</i><br>sudo !! <i>(repeat with sudo)</i>                     |                                                         |

| Task                      | CMD                                                                                                             | PowerShell                                                                                                                                      | Bash                                                                                                                                                        | Notes                                                                                    |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Search history            | F7 <i>(dialog)</i><br>type a prefix, then F8 <i>(cycle matches)</i>                                             | Ctrl+R <i>(reverse search)</i><br>type a prefix, then F8<br>Get-History   Where CommandLine -like '*git*'                                       | Ctrl+R <i>(reverse search)</i><br>history   grep pattern                                                                                                    |                                                                                          |
| Aliases                   | doskey ll=dir /B \$* <i>(this session)</i><br><i>(persist via a startup script in the AutoRun registry key)</i> | Set-Alias ll Get-ChildItem<br>Get-Alias <i>(list all)</i><br>function gs { git status } <i>(aliases cannot take arguments; use a function)</i>  | alias ll='ls -la'<br>alias <i>(list all)</i><br>unalias ll                                                                                                  |                                                                                          |
| Shell profile / startup   | HKCU\Software\Microsoft\Command Processor\AutoRun <i>(registry value)</i>                                       | \$PROFILE <i>(path to your profile script)</i><br>notepad \$PROFILE<br>. \$PROFILE <i>(reload)</i>                                              | ~/.bashrc <i>(interactive Bash)</i><br>~/.bash_profile <i>(login shells)</i><br>~/.zshrc <i>(zsh; default on macOS)</i><br>source ~/.bashrc <i>(reload)</i> | Put aliases, functions and PATH changes in the profile so they survive between sessions. |
| Exit code of last command | echo %ERRORLEVEL%<br>if %ERRORLEVEL% neq 0 echo failed                                                          | \$? <i>(True or False)</i><br>\$LASTEXITCODE <i>(from the last native .exe)</i>                                                                 | echo \$? <i>(0 means success)</i>                                                                                                                           |                                                                                          |
| Chain commands            | a && b <i>(b only if a succeeds)</i><br>a    b <i>(b only if a fails)</i><br>a & b <i>(both, regardless)</i>    | a; b <i>(both, regardless)</i><br>a && b, a    b <i>(PowerShell 7+ only)</i><br><i>(-and / -or are boolean operators, not chaining)</i>         | a && b<br>a    b<br>a; b                                                                                                                                    |                                                                                          |
| ★ Get help                | dir /?<br>help copy                                                                                             | Get-Help Get-Process -Examples<br>Get-Help *-Service <i>(search by name)</i><br>Get-Command -Noun Service<br>Update-Help <i>(once, 🛠 admin)</i> | man ls<br>ls --help<br>tldr ls <i>(needs install)</i>                                                                                                       |                                                                                          |
| Run as admin / root       | runas /user:Administrator cmd<br><i>(or right-click the shell and choose Run as administrator)</i>              | Start-Process powershell -Verb RunAs<br>sudo command <i>(Windows 11 24H2+; enable in Settings &gt; System &gt; For developers)</i>              | sudo command<br>sudo -i <i>(root shell)</i><br>su - <i>(switch user)</i>                                                                                    |                                                                                          |

## Text Processing and Output

| Task                  | CMD                                                          | PowerShell                                                                                                                                                                                  | Bash                                                                                    | Notes |
|-----------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|
| Display text          | echo Hello World<br>echo %PATH%<br>echo. <i>(blank line)</i> | Write-Output 'Hello' <i>(alias: echo, write)</i><br>Write-Host 'Text' -ForegroundColor Red<br><i>(console only, not pipeable)</i><br>"Value: \$var" <i>(double quotes expand variables)</i> | echo "Hello World"<br>printf "%s\n" "text"<br>echo -e "a\tb" <i>(interpret escapes)</i> |       |
| View a page at a time | more file.txt<br>dir /S   more                               | Get-Content file.txt   more<br>Get-Content file.txt   Out-Host -Paging                                                                                                                      | less file.txt <i>(q quits, / searches)</i><br>more file.txt<br>cat file.txt   less      |       |

| Task                     | CMD                                                                                                   | PowerShell                                                                                                                            | Bash                                                                                                                                                                     | Notes                                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| First lines of a file    | more +5 file.txt <i>(skip the first 5 instead)</i><br><i>(none; use PowerShell)</i>                   | Get-Content file.txt -Head 10<br>Get-Content file.txt   Select-Object -First 10                                                       | head file.txt <i>(10 lines)</i><br>head -n 20 file.txt<br>head -c 100 file.bin <i>(bytes)</i>                                                                            |                                                                                                                                   |
| Last lines of a file     | <i>(none; use PowerShell)</i>                                                                         | Get-Content file.txt -Tail 10<br>Get-Content log.txt -Wait <i>(follow, like tail -f)</i>                                              | tail file.txt<br>tail -n 20 file.txt<br>tail -f log.txt <i>(follow)</i>                                                                                                  |                                                                                                                                   |
| Count lines / words      | find /c /v "" < file.txt <i>(lines)</i><br>type file.txt   find /c "error"<br><i>(matching lines)</i> | Get-Content file.txt   Measure-Object -Line -Word -Character<br>(Get-Content file.txt).Count <i>(lines)</i>                           | wc -l file.txt <i>(lines)</i><br>wc -w file.txt <i>(words)</i><br>grep -c error file.txt <i>(matching lines)</i>                                                         |                                                                                                                                   |
| Sort lines               | sort file.txt<br>sort /R file.txt <i>(reverse)</i><br>dir /B   sort                                   | Get-Content file.txt   Sort-Object<br>Sort-Object -Unique<br>Get-Process   Sort-Object CPU -Descending                                | sort file.txt<br>sort -r <i>(reverse)</i> , -n <i>(numeric)</i> , -u <i>(unique)</i><br>sort -k2 file.txt <i>(by second field)</i>                                       |                                                                                                                                   |
| Unique lines             | <i>(none; use PowerShell)</i>                                                                         | Get-Content file.txt   Sort-Object -Unique<br>Get-Content file.txt   Group-Object   Sort Count -Desc <i>(with counts)</i>             | sort -u file.txt<br>sort file.txt   uniq -c <i>(with counts)</i><br>uniq file.txt <i>(adjacent duplicates only)</i>                                                      |                                                                                                                                   |
| Filter a stream          | dir   find "txt"<br>tasklist   findstr /I chrome                                                      | Get-Process   Where-Object Name -like 'chrome*' <i>(alias: where, ?)</i><br>Get-Process   Where CPU -gt 100<br>gc log.txt   sls error | ps aux   grep chrome<br>grep -v debug log.txt <i>(invert match)</i>                                                                                                      | PowerShell pipes objects, not text: filter on properties with Where-Object, then shape output with Select-Object or Format-Table. |
| Find and replace         | <i>(none; use PowerShell)</i>                                                                         | (Get-Content file.txt) -replace 'old','new'   Set-Content file.txt<br>'text' -replace '(\d+)', '\$1' <i>(regex by default)</i>        | sed 's/old/new/g' file.txt <i>(prints result)</i><br>sed -i 's/old/new/g' file.txt <i>(in place, Linux)</i><br>sed -i '' 's/old/new/g' file.txt <i>(in place, macOS)</i> | The parentheses around Get-Content read the whole file first so PowerShell can safely overwrite it.                               |
| Extract columns / fields | for /f "tokens=2 delims=," %i in (file.csv) do @echo %i<br><i>(inside a .bat file use %%i)</i>        | Import-Csv data.csv   Select-Object Name,Email<br>'a,b,c' -split ','<br>Get-Content f.txt   ForEach { (\$_ -split '\s+')[1] }         | cut -d, -f2 file.csv<br>awk '{print \$2}' file.txt<br>awk -F: '{print \$1}' /etc/passwd                                                                                  |                                                                                                                                   |

| Task                    | CMD                                                                                         | PowerShell                                                                                                                                                                                                                       | Bash                                                                                                                        | Notes                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Shape and export output | dir /B /O-D <i>(bare, newest first)</i><br>tasklist /FO CSV<br>tasklist /FO CSV > procs.csv | Get-Process   Select-Object Name,CPU<br>Get-Process   Format-Table -AutoSize<br>Get-Process   Format-List *<br>Get-Process   Export-Csv procs.csv -NoTypeInfoation<br>Get-Process   ConvertTo-Json<br>Get-Service   Out-GridView | column -t file.txt <i>(align columns)</i><br>ls -l   awk '{print \$9, \$5}'<br>jq . file.json <i>(needs install)</i>        | Format-* cmdlets must be last in a pipeline; their output is for the screen only. |
| Clipboard               | dir   clip                                                                                  | Get-Process   Set-Clipboard <i>(alias: scb)</i><br>Get-Clipboard <i>(alias: gcb)</i><br>ls   clip                                                                                                                                | ls   pbcopy (macOS), pbpaste<br>ls   xclip -selection clipboard <i>(Linux, X11)</i><br>ls   wl-copy <i>(Linux, Wayland)</i> |                                                                                   |

## Redirection and Piping

| Task                          | CMD                                                                     | PowerShell                                                                                                    | Bash                                                                                                     | Notes                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Redirect output (overwrite) > | dir > list.txt                                                          | Get-Process > processes.txt<br>Get-Process   Out-File -Encoding utf8 p.txt<br>Get-Process   Set-Content p.txt | ls -la > listing.txt                                                                                     | Windows PowerShell 5.1 writes UTF-16 with >. Use Out-File -Encoding utf8 for files other tools will read. PowerShell 7 defaults to UTF-8. |
| Redirect output (append) >>   | echo text >> file.txt                                                   | Get-Date >> log.txt<br>Add-Content log.txt 'line'                                                             | echo "new line" >> file.txt                                                                              |                                                                                                                                           |
| Redirect input <              | sort < unsorted.txt                                                     | Get-Content input.txt   Sort-Object<br><i>(PowerShell has no &lt; operator)</i>                               | sort < unsorted.txt<br>cat <<EOF ... EOF <i>(here-document)</i><br>cmd <<< "string" <i>(here-string)</i> |                                                                                                                                           |
| Pipe                          | dir   find "txt"<br>type file.txt   more                                | Get-Process   Sort-Object CPU -Descending   Select -First 5<br>Get-ChildItem   Where Length -gt 1MB           | ls -la   grep txt<br>cat file.txt   sort   uniq -c                                                       | CMD and Bash pipe text. PowerShell pipes objects; \$_ (or \$PSItem) is the current object inside a script block.                          |
| Redirect errors 2>            | command 2> errors.txt<br>command 2>&1 <i>(merge errors into output)</i> | command 2> errors.txt<br>command 2>&1<br>\$ErrorActionPreference = 'Stop' <i>(make errors terminate)</i>      | command 2> errors.txt<br>command 2>&1                                                                    |                                                                                                                                           |
| Redirect everything           | command > out.txt 2>&1                                                  | command *> all.txt<br>command 3>&1 (warnings), 4>&1 (verbose), 6>&1 <i>(information)</i>                      | command &> out.txt<br>command > out.txt 2>&1                                                             |                                                                                                                                           |

| Task                  | CMD                                      | PowerShell                                                                                                                           | Bash                                                                                                                  | Notes |
|-----------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------|
| Suppress output       | command > nul<br>command > nul 2>&1      | command   Out-Null<br>command > \$null<br>\$null = command <i>(fastest)</i>                                                          | command > /dev/null<br>command &> /dev/null<br>command 2> /dev/null <i>(errors only)</i>                              |       |
| Tee (file and screen) | <i>(none; use PowerShell)</i>            | Get-Process   Tee-Object -FilePath p.txt<br>Get-Process   Tee-Object -Variable procs   Select Name                                   | ls   tee listing.txt<br>ls   tee -a listing.txt <i>(append)</i><br>echo x   sudo tee /etc/file <i>(write as root)</i> |       |
| Command substitution  | for /f %i in ('date /T') do set today=%i | \$today = Get-Date -Format yyyy-MM-dd<br>Write-Output "Today: \$(Get-Date)"<br>Copy-Item a.txt "backup-\$(Get-Date -f yyyyMMdd).txt" | today=\$(date +%F)<br>echo "Today: \$(date)"<br>cp a.txt "backup-\$(date +%F).txt"                                    |       |

## Users and Groups

| Task              | CMD                                                              | PowerShell                                                                                   | Bash                                                                                                                                             | Notes                                                                                         |
|-------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| List users        | net user<br>net user alice <i>(details)</i>                      | Get-LocalUser<br>Get-LocalUser   Where Enabled                                               | cat /etc/passwd <i>(Linux)</i><br>getent passwd<br>dscl . list /Users <i>(macOS)</i>                                                             |                                                                                               |
| Add user          | net user alice * /add <i>(prompts for password)</i>              | New-LocalUser alice -Password (Read-Host -AsSecureString)<br>New-LocalUser alice -NoPassword | sudo adduser alice <i>(interactive, Debian/Ubuntu)</i><br>sudo useradd -m alice && sudo passwd alice                                             | 🔒 Requires admin or sudo. Avoid typing passwords on the command line: they end up in history. |
| Delete user       | net user alice /delete                                           | Remove-LocalUser alice                                                                       | sudo userdel alice<br>sudo userdel -r alice <i>(also remove home dir)</i>                                                                        | 🔒 Requires admin or sudo.                                                                     |
| Change password   | net user alice * <i>(prompts)</i>                                | Set-LocalUser alice -Password (Read-Host -AsSecureString)                                    | passwd <i>(your own)</i><br>sudo passwd alice                                                                                                    |                                                                                               |
| List groups       | net localgroup<br>net localgroup Administrators <i>(members)</i> | Get-LocalGroup<br>Get-LocalGroupMember Administrators                                        | groups <i>(current user)</i><br>getent group<br>id alice                                                                                         |                                                                                               |
| Add user to group | net localgroup Administrators alice /add                         | Add-LocalGroupMember -Group Administrators -Member alice                                     | sudo usermod -aG sudo alice <i>(Debian/Ubuntu)</i><br>sudo usermod -aG wheel alice <i>(Fedora/RHEL)</i><br><i>(log out and back in to apply)</i> | 🔒 Requires admin or sudo. ⚠ Forgetting -a in usermod removes the user from all other groups.  |

## Package Management

| Task              | CMD                                                                        | PowerShell                                                                                             | Bash                                                                                                                                                                | Notes                                                                                                                                                                 |
|-------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List installed    | winget list                                                                | winget list<br>choco list <i>(Chocolatey)</i><br>Get-Module -ListAvailable <i>(PowerShell modules)</i> | apt list --installed <i>(Debian/Ubuntu)</i><br>dnf list installed <i>(Fedora/RHEL)</i><br>brew list <i>(macOS)</i><br>pacman -Q <i>(Arch)</i>                       | wmic product get name is deprecated and very slow; use winget instead.                                                                                                |
| Search            | winget search vscode                                                       | winget search vscode<br>choco search vscode<br>Find-Module PSReadLine                                  | apt search vscode<br>dnf search vscode<br>brew search vscode                                                                                                        |                                                                                                                                                                       |
| Show package info | winget show Git.Git                                                        | winget show Git.Git                                                                                    | apt show git<br>dnf info git<br>brew info git                                                                                                                       |                                                                                                                                                                       |
| Install           | winget install Git.Git<br>winget install --id Git.Git -e <i>(exact id)</i> | winget install Git.Git<br>choco install git -y<br>Install-Module PSReadLine <i>(PowerShell module)</i> | sudo apt install git<br>sudo dnf install git<br>brew install git<br>sudo pacman -S git                                                                              |  Usually needs admin or sudo. brew and per-user winget installs are the exception. |
| Update            | winget upgrade <i>(list available)</i><br>winget upgrade --all             | winget upgrade --all<br>choco upgrade all<br>Update-Module                                             | sudo apt update && sudo apt upgrade<br>sudo dnf upgrade<br>brew update && brew upgrade<br>sudo pacman -Syu                                                          | apt update only refreshes the package index. apt upgrade does the installing.                                                                                         |
| Remove            | winget uninstall Git.Git                                                   | winget uninstall Git.Git<br>choco uninstall git<br>Uninstall-Module name                               | sudo apt remove git <i>(keeps config)</i><br>sudo apt purge git <i>(removes config)</i><br>sudo apt autoremove <i>(orphaned dependencies)</i><br>brew uninstall git |                                                                                                                                                                       |

## Developer Setup

| Task                                  | Windows                                                                                                                                                                                                                                                                                                                                                                                                         | macOS                                                                                                                                                                       | Linux                                                                                                                                                                           | Notes                                                                                                                                                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Install WSL (Linux on Windows)</b> | <code>wsl --install</code> (Ubuntu by default; reboot afterwards)<br><code>wsl --install -d Debian</code> (pick a distro)<br><code>wsl -l -v</code> (list distros and versions)<br><code>wsl --update</code><br><code>wsl --shutdown</code><br><code>wsl</code> (open the default distro)                                                                                                                       | (not needed; macOS is already Unix)                                                                                                                                         | (not needed)                                                                                                                                                                    |  Run from an administrator prompt. Inside WSL, Windows drives are under /mnt/c and Windows tools are on the PATH, so <code>code</code> opens VS Code. |
| <b>Package manager</b>                | <code>winget --version</code> (built in on Windows 10 1709+ and 11)<br><code>winget source update</code><br><i>(Chocolatey is optional: see <a href="https://chocolatey.org/install">chocolatey.org/install</a>)</i>                                                                                                                                                                                            | <code>/bin/bash -c "\$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"</code><br><code>brew doctor</code>                                   | <i>(apt, dnf or pacman is already installed)</i><br><code>sudo apt update</code> (refresh the index first)                                                                      | If winget is missing, install "App Installer" from the Microsoft Store.                                                                                                                                                                  |
| <b>Git</b>                            | <code>winget install --id Git.Git -e git --version</code>                                                                                                                                                                                                                                                                                                                                                       | <code>xcode-select --install</code> (Apple's git plus compilers)<br><code>brew install git</code> (newer version)                                                           | <code>sudo apt install git</code><br><code>sudo dnf install git</code>                                                                                                          | On Windows this also installs Git Bash.                                                                                                                                                                                                  |
| <b>GitHub CLI</b>                     | <code>winget install --id GitHub.cli -e gh auth login</code>                                                                                                                                                                                                                                                                                                                                                    | <code>brew install gh</code><br><code>gh auth login</code>                                                                                                                  | <code>sudo apt install gh</code> (Ubuntu 23.04+; older distros: see <a href="https://cli.github.com">cli.github.com</a> )<br><code>gh auth login</code>                         | gh auth login signs in through the browser and stores credentials that git push uses too.                                                                                                                                                |
| <b>SSH key for GitHub</b>             | <code>ssh-keygen -t ed25519 -C "you@example.com"</code><br><code>Get-Service ssh-agent   Set-Service -StartupType Automatic; Start-Service ssh-agent</code> (PowerShell,  admin)<br><code>ssh-add</code><br><code>\$env:USERPROFILE\.ssh\id_ed25519</code> (PowerShell)<br><code>gh ssh-key add ~/.ssh/id_ed25519.pub</code> | <code>ssh-keygen -t ed25519 -C "you@example.com"</code><br><code>ssh-add --apple-use-keychain ~/.ssh/id_ed25519</code><br><code>gh ssh-key add ~/.ssh/id_ed25519.pub</code> | <code>ssh-keygen -t ed25519 -C "you@example.com"</code><br><code>eval "\$(ssh-agent -s)"; ssh-add ~/.ssh/id_ed25519</code><br><code>gh ssh-key add ~/.ssh/id_ed25519.pub</code> | Test with <code>ssh -T git@github.com</code> . The Windows ssh-agent service is off by default.                                                                                                                                          |
| <b>Node.js</b>                        | <code>winget install OpenJS.NodeJS.LTS</code><br><code>winget install Schniz.fnm</code> (version manager)<br><code>node -v; npm -v</code>                                                                                                                                                                                                                                                                       | <code>brew install node</code><br><code>brew install fnm</code> (version manager)                                                                                           | <code>curl -fsSL https://fnm.vercel.app/install   bash</code> (version manager)<br><code>fnm install --lts</code><br><i>(distro packages are often years old)</i>               | A version manager such as <code>fnm</code> or <code>nvm</code> lets you switch Node versions per project.                                                                                                                                |
| <b>Python</b>                         | <code>winget install Python.Python.3.12</code><br><code>python --version</code><br><code>py -3</code> (launcher; picks an installed version)                                                                                                                                                                                                                                                                    | <code>brew install python</code><br><code>python3 --version</code>                                                                                                          | <code>sudo apt install python3 python3-pip python3-venv</code><br><code>python3 --version</code>                                                                                | Per-project environment: <code>python -m venv .venv</code> , then <code>.venv\Scripts\activate</code> (Windows) or <code>source .venv/bin/activate</code> .                                                                              |

| Task                                     | Windows                                                                                                                  | macOS                                                                                                                    | Linux                                                                                                                           | Notes                                                                                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VS Code</b>                           | winget install Microsoft.VisualStudioCode<br>code . <i>(open the current folder)</i>                                     | brew install --cask visual-studio-code<br>code . <i>(first run Shell Command: Install code from the Command Palette)</i> | sudo snap install code --classic<br><i>(or the .deb / .rpm from code.visualstudio.com)</i><br>code .                            |                                                                                                                                                                        |
| <b>PowerShell 7 and Windows Terminal</b> | winget install Microsoft.PowerShell<br>winget install Microsoft.WindowsTerminal<br>pwsh <i>(start PowerShell 7)</i>      | brew install --cask powershell<br>pwsh                                                                                   | <i>(see Microsoft's install page for your distro)</i><br>pwsh                                                                   | Windows PowerShell 5.1 is built in. PowerShell 7 (pwsh) is the current cross-platform version and installs alongside it. Windows 11 already includes Windows Terminal. |
| <b>Compilers and build tools</b>         | winget install Microsoft.VisualStudio.2022.BuildTools<br><i>(choose "Desktop development with C++" in the installer)</i> | xcode-select --install                                                                                                   | sudo apt install build-essential<br>sudo dnf groupinstall "Development Tools"                                                   | Needed by npm and pip packages that compile native code.                                                                                                               |
| <b>Docker</b>                            | winget install Docker.DockerDesktop<br>docker run hello-world                                                            | brew install --cask docker<br>docker run hello-world                                                                     | curl -fsSL https://get.docker.com   sh<br>sudo usermod -aG docker \$USER <i>(then log out and in)</i><br>docker run hello-world | Docker Desktop on Windows runs on WSL 2, so install WSL first.                                                                                                         |
| <b>Allow PowerShell scripts</b>          | Set-ExecutionPolicy RemoteSigned -Scope CurrentUser <i>(PowerShell)</i><br>Get-ExecutionPolicy -List                     | <i>(not needed)</i>                                                                                                      | <i>(not needed)</i>                                                                                                             | Fixes "running scripts is disabled on this system". RemoteSigned runs local scripts and signed downloads only.                                                         |
| <b>Check what is installed</b>           | winget list<br>git --version; node -v; python --version; code -v<br>where git <i>(path to the executable)</i>            | brew list<br>git --version; node -v; python3 --version<br>which git                                                      | apt list --installed 2>/dev/null   grep -i git<br>git --version; node -v; python3 --version<br>which git                        | After installing anything, open a new terminal window. Existing shells keep the old PATH.                                                                              |

## Git

| Task                      | Command                                                                                                                                                                                      | Notes                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>One-time setup</b>     | git config --global user.name "Your Name"<br>git config --global user.email "you@example.com"<br>git config --global init.defaultBranch main<br>git config --list <i>(show all settings)</i> | Settings live in ~/.gitconfig. Drop --global to set them for one repository only. |
| <b>Start a repository</b> | git init<br>git clone https://github.com/user/repo.git<br>git clone --depth 1 URL <i>(latest commit only, fast)</i>                                                                          |                                                                                   |

| Task               | Command                                                                                                                                                                                                                                                                   | Notes                                                                                                                  |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| See what changed   | <pre>git status git status -s <i>(short form)</i> git diff <i>(unstaged changes)</i> git diff --staged <i>(what will be committed)</i> git diff main..feature <i>(between branches)</i></pre>                                                                             |                                                                                                                        |
| History            | <pre>git log --oneline --graph --all git log -p file.txt <i>(changes to one file)</i> git log -5 --stat git show HEAD <i>(last commit in full)</i> git blame file.txt <i>(who changed each line)</i></pre>                                                                |                                                                                                                        |
| Stage and commit   | <pre>git add file.txt git add -A <i>(everything)</i> git add -p <i>(pick hunks interactively)</i> git commit -m "Message" git commit -am "Message" <i>(add tracked files and commit)</i> git commit --amend <i>(fix the last commit)</i></pre>                            | ⚠ Do not amend a commit you have already pushed; it rewrites history others may have.                                  |
| Unstage or discard | <pre>git restore --staged file.txt <i>(unstage, keep changes)</i> git restore file.txt <i>(discard local changes)</i> git clean -fd <i>(delete untracked files and folders)</i> git clean -n <i>(dry run first)</i></pre>                                                 | ⚠ restore and clean throw work away with no undo. Run git clean -n first.                                              |
| Branches           | <pre>git branch <i>(list)</i> git branch -a <i>(include remote)</i> git switch -c feature <i>(create and switch)</i> git switch main git branch -m old new <i>(rename)</i> git branch -d feature <i>(delete merged)</i> git branch -D feature <i>(force delete)</i></pre> | switch and restore replaced the overloaded checkout in Git 2.23. checkout still works everywhere.                      |
| Merge and rebase   | <pre>git merge feature <i>(merge into current branch)</i> git rebase main <i>(replay current branch onto main)</i> git rebase -i HEAD~3 <i>(squash or reorder last 3)</i> git merge --abort / git rebase --abort</pre>                                                    | Merge keeps history as it happened. Rebase makes it linear. ⚠ Never rebase commits that are already pushed and shared. |
| Remotes            | <pre>git remote -v git remote add origin https://github.com/user/repo.git git remote set-url origin NEW_URL</pre>                                                                                                                                                         |                                                                                                                        |

| Task                     | Command                                                                                                                                                                                                                                                                                                             | Notes                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Push and pull            | git push -u origin main <i>(first push; sets upstream)</i><br>git push<br>git pull <i>(fetch and merge)</i><br>git pull --rebase<br>git fetch <i>(download without merging)</i><br>git push origin --delete feature <i>(delete remote branch)</i>                                                                   |                                                                                                                                   |
| Undo commits             | git revert HEAD <i>(new commit that undoes the last one; safe)</i><br>git reset --soft HEAD~1 <i>(uncommit, keep changes staged)</i><br>git reset --mixed HEAD~1 <i>(uncommit, keep changes unstaged)</i><br>git reset --hard HEAD~1 <i>(uncommit and discard changes)</i><br>git reflog <i>(find lost commits)</i> | ⚠ reset --hard deletes work. revert is the safe choice for anything already pushed. reflog can recover a reset for about 30 days. |
| Stash                    | git stash <i>(shelve changes)</i><br>git stash push -m "wip" <i>(with a name)</i><br>git stash list<br>git stash pop <i>(restore latest and drop it)</i><br>git stash apply stash@{1}<br>git stash drop                                                                                                             |                                                                                                                                   |
| Tags and releases        | git tag <i>(list)</i><br>git tag -a v1.0 -m "Release 1.0"<br>git push --tags<br>git checkout v1.0 <i>(look at a tag)</i>                                                                                                                                                                                            |                                                                                                                                   |
| Ignore and untrack files | git rm --cached file.txt <i>(stop tracking, keep file)</i><br>git rm -r --cached folder/<br>git check-ignore -v file.txt <i>(why is this ignored?)</i><br>git mv old.txt new.txt <i>(rename and stage)</i>                                                                                                          | Patterns in .gitignore only affect files not yet tracked. Untrack committed files with git rm --cached first.                     |
| Search                   | git grep "pattern" <i>(search tracked files)</i><br>git log -S "text" <i>(commits that added or removed text)</i><br>git log --grep "fix" <i>(search commit messages)</i><br>git log --author="Name"                                                                                                                |                                                                                                                                   |
| Who did what             | git shortlog -sn <i>(commits per author)</i><br>git log --since="2 weeks ago"<br>git bisect start / git bisect good v1.0 / git bisect bad <i>(find the breaking commit)</i>                                                                                                                                         |                                                                                                                                   |
| Help                     | git help commit<br>git commit -h <i>(short flag list)</i><br>git <command> --help                                                                                                                                                                                                                                   |                                                                                                                                   |

# Docker

| Task                   | Command                                                                                                                                                                                                                                                                                                                    | Notes                                                                                                                                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| List containers        | <code>docker ps</code> <i>(running)</i><br><code>docker ps -a</code> <i>(include stopped)</i><br><code>docker ps -q</code> <i>(IDs only)</i>                                                                                                                                                                               | On Linux, prefix with <code>sudo</code> or add yourself to the docker group. Docker Desktop on Windows and macOS runs the daemon for you. |
| List images            | <code>docker images</code><br><code>docker image ls</code><br><code>docker history nginx</code> <i>(layers)</i>                                                                                                                                                                                                            |                                                                                                                                           |
| Run a container        | <code>docker run -d -p 8080:80 --name web nginx</code> <i>(detached, port map, named)</i><br><code>docker run -it ubuntu bash</code> <i>(interactive shell)</i><br><code>docker run --rm alpine echo hi</code> <i>(remove when it exits)</i><br><code>docker run -v \${PWD}:/app node</code> <i>(mount current folder)</i> | -d runs in the background, -p maps host:container ports, -v mounts a volume, --rm cleans up on exit.                                      |
| Shell into a container | <code>docker exec -it web bash</code><br><code>docker exec -it web sh</code> <i>(if bash is missing)</i><br><code>docker attach web</code> <i>(attach to PID 1; Ctrl+P Ctrl+Q to detach)</i>                                                                                                                               |                                                                                                                                           |
| Logs                   | <code>docker logs web</code><br><code>docker logs -f web</code> <i>(follow)</i><br><code>docker logs --tail 50 web</code>                                                                                                                                                                                                  |                                                                                                                                           |
| Stop, start, remove    | <code>docker stop web</code><br><code>docker start web</code><br><code>docker restart web</code><br><code>docker rm web</code> <i>(add -f to force)</i><br><code>docker rm -f \$(docker ps -aq)</code> <i>(remove all ⚠)</i>                                                                                               | ⚠ The last command removes every container.                                                                                               |
| Remove images          | <code>docker rmi nginx</code><br><code>docker image rm nginx:latest</code><br><code>docker image prune</code> <i>(dangling)</i><br><code>docker image prune -a</code> <i>(all unused)</i>                                                                                                                                  |                                                                                                                                           |
| Free up space          | <code>docker system df</code> <i>(show usage)</i><br><code>docker system prune</code> <i>(stopped containers, unused networks, dangling images)</i><br><code>docker system prune -a --volumes</code> <i>(⚠ everything unused, including volumes)</i>                                                                       | ⚠ <code>prune -a --volumes</code> deletes data in unused volumes. Check docker volume ls first.                                           |
| Build an image         | <code>docker build -t myapp .</code><br><code>docker build -t myapp:1.0 -f Dockerfile.prod .</code><br><code>docker tag myapp myrepo/myapp:1.0</code>                                                                                                                                                                      | The trailing dot is the build context, the folder sent to the daemon.                                                                     |

| Task                 | Command                                                                                                                                                                                                | Notes                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Push and pull        | docker pull redis:7<br>docker login<br>docker push myrepo/myapp:1.0                                                                                                                                    |                                                                                                                                              |
| Copy files           | docker cp web:/etc/nginx/nginx.conf ./nginx.conf <i>(out of the container)</i><br>docker cp ./file.txt web:/tmp/ <i>(into the container)</i>                                                           |                                                                                                                                              |
| Inspect and stats    | docker inspect web <i>(full JSON)</i><br>docker stats <i>(live CPU and memory)</i><br>docker port web<br>docker top web <i>(processes)</i>                                                             |                                                                                                                                              |
| Volumes and networks | docker volume ls<br>docker volume create data<br>docker network ls<br>docker network create mynet                                                                                                      |                                                                                                                                              |
| Docker Compose       | docker compose up -d <i>(start in background)</i><br>docker compose down <i>(stop and remove)</i><br>docker compose ps<br>docker compose logs -f<br>docker compose build<br>docker compose exec web sh | docker compose (v2, a subcommand) replaces the old docker-compose script. It reads compose.yaml or docker-compose.yml in the current folder. |

## Keyboard Shortcuts

| Action                  | CMD                                                 | PowerShell                                              | Bash                                                         | Notes                                                                 |
|-------------------------|-----------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| Auto-complete           | Tab                                                 | Tab <i>(Ctrl+Space shows a menu)</i>                    | Tab <i>(press twice to list options)</i>                     |                                                                       |
| Previous / next command | ↑ / ↓ <i>(F3 repeats the last one)</i>              | ↑ / ↓                                                   | ↑ / ↓ or Ctrl+P / Ctrl+N                                     |                                                                       |
| Search history          | F7 <i>(list)</i> , F8 <i>(cycle prefix matches)</i> | Ctrl+R <i>(backward)</i> , Ctrl+S <i>(forward)</i> , F8 | Ctrl+R <i>(backward)</i> , Ctrl+S <i>(forward)</i>           |                                                                       |
| Clear screen            | cls <i>(no shortcut)</i>                            | Ctrl+L                                                  | Ctrl+L                                                       |                                                                       |
| Cancel running command  | Ctrl+C                                              | Ctrl+C                                                  | Ctrl+C                                                       |                                                                       |
| Clear current line      | Esc                                                 | Esc                                                     | Ctrl+U <i>(before cursor)</i> , Ctrl+K <i>(after cursor)</i> | In PowerShell, Ctrl+U and Ctrl+K also work with the default key mode. |

| Action                    | CMD                          | PowerShell                               | Bash                                                  | Notes                                                                                                                   |
|---------------------------|------------------------------|------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Start / end of line       | Home / End                   | Home / End <i>(also Ctrl+A / Ctrl+E)</i> | Ctrl+A / Ctrl+E <i>(Home / End in most terminals)</i> |                                                                                                                         |
| Move by word              | Ctrl+← / Ctrl+→              | Ctrl+← / Ctrl+→                          | Alt+B / Alt+F <i>(Ctrl+← / → in many terminals)</i>   |                                                                                                                         |
| Delete word before cursor | <i>(none)</i>                | Ctrl+Backspace                           | Ctrl+W                                                |                                                                                                                         |
| Delete word after cursor  | <i>(none)</i>                | Ctrl+Delete                              | Alt+D                                                 |                                                                                                                         |
| Undo an edit              | <i>(none)</i>                | Ctrl+Z                                   | Ctrl+_ or Ctrl+X Ctrl+U                               |                                                                                                                         |
| Paste                     | Ctrl+V or right-click        | Ctrl+V or right-click                    | Ctrl+Shift+V <i>(Cmd+V on macOS)</i>                  | Copy and paste are handled by the terminal app, not the shell. In Bash Ctrl+C cancels and Ctrl+V inserts a literal key. |
| Copy                      | Select, then Enter or Ctrl+C | Select, then Enter or Ctrl+C             | Ctrl+Shift+C <i>(Cmd+C on macOS)</i>                  |                                                                                                                         |
| Pause / resume output     | Ctrl+S / Ctrl+Q              | Ctrl+S / Ctrl+Q                          | Ctrl+S / Ctrl+Q                                       | If the terminal seems frozen, you probably hit Ctrl+S. Press Ctrl+Q.                                                    |
| Suspend foreground job    | <i>(none)</i>                | <i>(none)</i>                            | Ctrl+Z <i>(resume with fg or bg)</i>                  |                                                                                                                         |
| Exit shell                | exit                         | exit <i>(Ctrl+D on an empty line)</i>    | exit or Ctrl+D                                        | In CMD, Ctrl+Z is the end-of-file character, not exit.                                                                  |

## Notes and Gotchas

### Paths

- Windows uses backslashes (C:\Users\Alice). Bash uses forward slashes (/home/alice). PowerShell accepts both.
- Quote any path containing spaces: "C:\Program Files" or '/My Files'.
- Windows file systems are case-insensitive. Linux is case-sensitive: File.txt and file.txt are different files. macOS is case-insensitive by default.

### Quoting and escaping

- CMD: escape special characters (& < > | ^) with a caret: echo a ^& b. Variables are %NAME%.
- PowerShell: single quotes are literal, double quotes expand \$variables and \$(expressions). The escape character is the backtick (`).
- Bash: single quotes are literal, double quotes expand \$variables and \$(commands). The escape character is the backslash (\).

### Wildcards

- \* matches any number of characters (\*.txt). ? matches exactly one character (file?.txt).
- [abc] and [a-z] match one character from a set (Bash and PowerShell, not CMD).
- In Bash the shell expands wildcards before the command runs. Quote them ("\*.txt") when the command should see the pattern itself, as with find and grep.

## PowerShell is different

- Commands are Verb-Noun cmdlets (Get-Process, Stop-Service). Aliases such as ls, cat, rm and ps map to cmdlets, but their flags are PowerShell flags: ls -Force, not ls -a.
- Pipelines carry objects, not text. Filter with Where-Object, pick columns with Select-Object, sort with Sort-Object, and format last.
- Windows PowerShell 5.1 ships with Windows. PowerShell 7 (pwsh) is a separate install with && and | |, UTF-8 defaults and Get-Uptime.

## macOS, zsh and WSL

- macOS uses zsh as its default shell. Nearly everything in the Bash column works unchanged; profile files are ~/.zshrc instead of ~/.bashrc.
- macOS ships BSD versions of many tools, so some Linux flags differ (sed -i, ls --color, du --max-depth). Commands marked (Linux) do not exist on macOS and vice versa.
- Windows Subsystem for Linux (wsl) gives you a real Linux shell on Windows. Windows drives appear under /mnt/c.

## Setting up Windows for development

- Turn on Developer Mode (Settings > System > For developers). It lets you create symlinks without an admin prompt and, on Windows 11 24H2+, enables the built-in sudo.
- Long paths: node\_modules trees often exceed the 260-character limit. Run git config --global core.longpaths true, and enable "Enable Win32 long paths" in Group Policy or set LongPathsEnabled=1 under HKLM\SYSTEM\CurrentControlSet\Control\FileSystem (🔒 admin).
- winget installs per user by default; add --scope machine (🔒 admin) to install for everyone.
- Most installers add themselves to the PATH, but only new terminal windows see it. If a command is "not recognized" right after installing, open a new window first.

## Git on Windows

- Git for Windows installs Git Bash, a Bash shell where the whole Bash column of this sheet works. Git itself works identically in CMD and PowerShell.
- Line endings: set git config --global core.autocrlf true on Windows (or add a .gitattributes with \*text=auto to the repo) so files do not flip between CRLF and LF.
- Credentials: Git for Windows includes Git Credential Manager, which opens a browser sign-in on the first push and remembers it. On macOS the keychain does the same; on Linux use git config --global credential.helper store or the GitHub CLI (gh auth login).
- Lots of "modified" files right after a clone with no real changes usually means a line-ending or file-mode mismatch: git config core.fileMode false and check core.autocrlf.